

Questions with Answer Keys

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Q1: 16 March (Shift 2) - Numerical

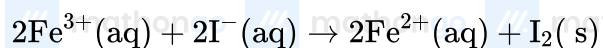
A 5.0 mol dm^{-3} aqueous solution of KCl has a conductance of 0.55 mS when measured in a cell constant 1.3 cm^{-1} . The molar conductivity of this solution is $\text{mS m}^2 \text{ mol}^{-1}$. (Round off to the Nearest Integer)

Q2: 17 March (Shift 2) - Numerical

A KC solution of conductivity 0.14 S m^{-1} shows a resistance of 4.19Ω in a conductivity cell. If the same cell is filled with an HCl solution, the resistance drops to 1.03Ω . The conductivity of the HCl solution is $\times 10^{-2} \text{ S m}^{-1}$. (Round off to the Nearest Integer).

Q3: 18 March (Shift 1) - Numerical

For the reaction



the magnitude of the standard molar free energy change, $\Delta_r G_m^\circ = -\text{kJ}$ (Round off to the Nearest Integer).

Q4: 18 March (Shift 2) - Numerical

The molar conductivities at infinite dilution of barium chloride, sulphuric acid and hydrochloric acid are 280 , 860 and $426 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. The molar conductivity at infinite dilution of barium sulphate is $\text{S cm}^2 \text{ mol}^{-1}$ (Round off to the Nearest Integer).

Answer Key**Q1 (14.3)****Q2 (57)****Q3 (45)****Q4 (288)**

Q1 (14.3)

Given concⁿ of KCl = $\frac{\text{m}\cdot\text{mol}}{\text{L}}$ $\therefore$ Conductance (G) = 0.55 mSCell constant $\left(\frac{\ell}{A}\right) = 1.3 \text{ cm}^{-1}$ To Calculate : Molar conductivity (λ_m) of sol. $\rightarrow$ $\rightarrow$ Molarity = $5 \times 10^{-3} \frac{\text{mol}}{\text{L}}$ $\rightarrow$ Conductivity = $G \times \left(\frac{\ell}{A}\right) = 0.55 \text{ mS} \times \frac{1.3}{100} \text{ m}^{-1}$ $= 55 \times 1.3 \text{ mSm}^{-1}$ $\text{eq}^n(1) \lambda_m = \frac{1}{1000} \times \frac{55 \times 1.3 \text{ mSm}^2}{\left(\frac{5}{1000}\right) \frac{\text{mol}}{\text{mol}}}$ $\Rightarrow \lambda_m = 14.3 \frac{\text{mSm}^2}{\text{mol}}$

Q2 (57)

 $\kappa = \frac{1}{R} \cdot G^*$ For same conductivity cell, G^* is constant and hence $\kappa \cdot R = \text{constant}$. $\therefore 0.14 \times 4.19 = \kappa \times 1.03$ or, $\kappa = \frac{0.14 \times 4.19}{1.03}$ $= 0.5695 \text{ Sm}^{-1}$ $= 56.95 \times 10^{-2} \text{ Sm}^{-1} \approx 57 \times 10^{-2} \text{ Sm}^{-1}$

Q3 (45)

Hints and Solutions

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$$E_1^0 + 2E_2^0 = 3E_3^0$$

$$E_1^0 = 3E_3^0 - 2E_2^0$$

$$= 3(-0.036) - 2(-0.44)$$

$$= +0.772 \text{ V}$$

$$E_{\text{cell}}^0 = E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 + E_{\text{I}^-/\text{I}_2}^0 = 0.233$$

$$\Delta_r G^0 = -2 \times 96.5 \times 0.233 = -45 \text{ kJ}$$

Q4 (288)

From Kohlrausch's law

$$\Lambda_m^\infty (\text{BaSO}_4) = \lambda_m^\infty (\text{Ba}^{2+}) + \lambda_m^\infty (\text{SO}_4^{2-})$$

$$\Lambda_m^\infty (\text{BaSO}_4) = \Lambda_m^\infty (\text{BaCl}_2) + \Lambda_m^\infty (\text{H}_2\text{SO}_4)$$

$$= 280 + 860 - 2(426)$$

$$= 288 \text{ Scm}^2 \text{ mol}^{-1}$$